

```
In [2]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Step 1: Load dataset
df = pd.read_csv("Sample - Superstore.csv", encoding="latin1")

print("Dataset loaded successfully!")

print("\nFirst 5 Records:")
print(df.head())

# Step 2: Check dataset
print("\nDataset Shape:")
print(df.shape)

print("\nDataset Information:")
df.info()

print("\nMissing Values:")
print(df.isnull().sum())

# Step 3: Bar Chart - Sales by Category
plt.figure(figsize=(8, 5))

sns.barplot(
    data=df,
    x="Category",
    y="Sales",
    estimator="sum"
)

plt.title("Total Sales by Category")
plt.xlabel("Category")
plt.ylabel("Total Sales")

plt.tight_layout()
plt.show()

# Step 4: Convert Order Date
# format="mixed" handles different date formats in the dataset

df["Order Date"] = pd.to_datetime(
    df["Order Date"],
    format="mixed"
)

# Sales by Year
year_data = df.groupby(
    df["Order Date"].dt.year
)["Sales"].sum()

# Step 5: Line Chart - Sales by Year
```

```
plt.figure(figsize=(10, 5))

plt.plot(
    year_data.index,
    year_data.values,
    marker="o"
)

plt.title("Total Sales by Year")
plt.xlabel("Year")
plt.ylabel("Total Sales")

plt.grid(True)

plt.tight_layout()
plt.show()

# Step 6: Histogram - Sales Distribution
plt.figure(figsize=(8, 5))

plt.hist(
    df["Sales"],
    bins=20
)

plt.title("Distribution of Sales")
plt.xlabel("Sales")
plt.ylabel("Frequency")

plt.tight_layout()
plt.show()

# Step 7: Box Plot - Profit by Category
plt.figure(figsize=(8, 5))

sns.boxplot(
    data=df,
    x="Category",
    y="Profit"
)

plt.title("Profit Distribution by Category")
plt.xlabel("Category")
plt.ylabel("Profit")

plt.tight_layout()
plt.show()

# Step 8: Scatter Plot - Sales vs Profit
plt.figure(figsize=(9, 5))

sns.scatterplot(
    data=df,
    x="Sales",
    y="Profit",
    hue="Category"
)
```

```
plt.title("Sales vs Profit")
plt.xlabel("Sales")
plt.ylabel("Profit")

plt.tight_layout()
plt.show()

# Step 9: Correlation Heatmap
numeric_data = df.select_dtypes(
    include="number"
)

correlation = numeric_data.corr()

plt.figure(figsize=(10, 7))

sns.heatmap(
    correlation,
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)

plt.title("Correlation Heatmap")

plt.tight_layout()
plt.show()

# Step 10: Customized Visualization - Sales by Region
region_sales = df.groupby(
    "Region"
)[ "Sales" ].sum().sort_values(ascending=False)

plt.figure(figsize=(8, 5))

bars = plt.bar(
    region_sales.index,
    region_sales.values
)

plt.title("Sales Distribution by Region")
plt.xlabel("Region")
plt.ylabel("Total Sales")

# Display values on bars
for bar in bars:

    height = bar.get_height()

    plt.text(
        bar.get_x() + bar.get_width() / 2,
        height,
        f"{height:.0f}",
        ha="center",
        va="bottom"
    )
```

```
plt.tight_layout()
plt.show()

# Step 11: Visualization Summary
print("\nVisualization Summary")
print("-" * 40)

print("1. The bar chart shows total sales for each product category.")

print("2. The line chart shows sales trends across different years.")

print("3. The histogram shows the distribution of sales values.")

print("4. The box plot compares profit distribution across categories.")

print("5. The scatter plot shows the relationship between sales and profit.")

print("6. The heatmap shows correlations between numerical variables.")

print("7. The customized chart compares sales across different regions.")

# Step 12: Business Insights
print("\nBusiness Insights")
print("-" * 40)

print(
    "Total Sales:",
    round(df["Sales"].sum(), 2)
)

print(
    "Total Profit:",
    round(df["Profit"].sum(), 2)
)

print(
    "Total Orders:",
    df["Order ID"].nunique()
)

print(
    "Total Customers:",
    df["Customer ID"].nunique()
)

print(
    "Best Category:",
    df.groupby("Category")["Sales"].sum().idxmax()
)

print(
    "Best Region:",
    df.groupby("Region")["Sales"].sum().idxmax()
)
```

```
# Step 13: Completion Message  
print("\nExperiment completed successfully!")
```

Dataset loaded successfully!

First 5 Records:

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	\
0	1	CA-2016-152156	11-08-2016	11-11-2016	Second Class	CG-12520	
1	2	CA-2016-152156	11-08-2016	11-11-2016	Second Class	CG-12520	
2	3	CA-2016-138688	06-12-2016	6/16/2016	Second Class	DV-13045	
3	4	US-2015-108966	10-11-2015	10/18/2015	Standard Class	SO-20335	
4	5	US-2015-108966	10-11-2015	10/18/2015	Standard Class	SO-20335	

	Customer Name	Segment	Country	City	...	\
0	Claire Gute	Consumer	United States	Henderson	...	
1	Claire Gute	Consumer	United States	Henderson	...	
2	Darrin Van Huff	Corporate	United States	Los Angeles	...	
3	Sean O'Donnell	Consumer	United States	Fort Lauderdale	...	
4	Sean O'Donnell	Consumer	United States	Fort Lauderdale	...	

	Postal Code	Region	Product ID	Category	Sub-Category	\
0	42420	South	FUR-BO-10001798	Furniture	Bookcases	
1	42420	South	FUR-CH-10000454	Furniture	Chairs	
2	90036	West	OFF-LA-10000240	Office Supplies	Labels	
3	33311	South	FUR-TA-10000577	Furniture	Tables	
4	33311	South	OFF-ST-10000760	Office Supplies	Storage	

	Product Name	Sales	Quantity	\
0	Bush Somerset Collection Bookcase	261.9600	2	
1	Hon Deluxe Fabric Upholstered Stacking Chairs,...	731.9400	3	
2	Self-Adhesive Address Labels for Typewriters b...	14.6200	2	
3	Bretford CR4500 Series Slim Rectangular Table	957.5775	5	
4	Eldon Fold 'N Roll Cart System	22.3680	2	

	Discount	Profit
0	0.00	41.9136
1	0.00	219.5820
2	0.00	6.8714
3	0.45	-383.0310
4	0.20	2.5164

[5 rows x 21 columns]

Dataset Shape:

(9994, 21)

Dataset Information:

```
<class 'pandas.core.frame.DataFrame'>
```

RangeIndex: 9994 entries, 0 to 9993

Data columns (total 21 columns):

#	Column	Non-Null Count	Dtype
0	Row ID	9994 non-null	int64
1	Order ID	9994 non-null	object
2	Order Date	9994 non-null	object
3	Ship Date	9994 non-null	object
4	Ship Mode	9994 non-null	object
5	Customer ID	9994 non-null	object
6	Customer Name	9994 non-null	object
7	Segment	9994 non-null	object
8	Country	9994 non-null	object
9	City	9994 non-null	object
10	State	9994 non-null	object

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11 Postal Code      9994 non-null    int64
12 Region           9994 non-null    object
13 Product ID       9994 non-null    object
14 Category         9994 non-null    object
15 Sub-Category     9994 non-null    object
16 Product Name     9994 non-null    object
17 Sales            9994 non-null    float64
18 Quantity         9994 non-null    int64
19 Discount         9994 non-null    float64
20 Profit           9994 non-null    float64

```

dtypes: float64(3), int64(3), object(15)

memory usage: 1.6+ MB

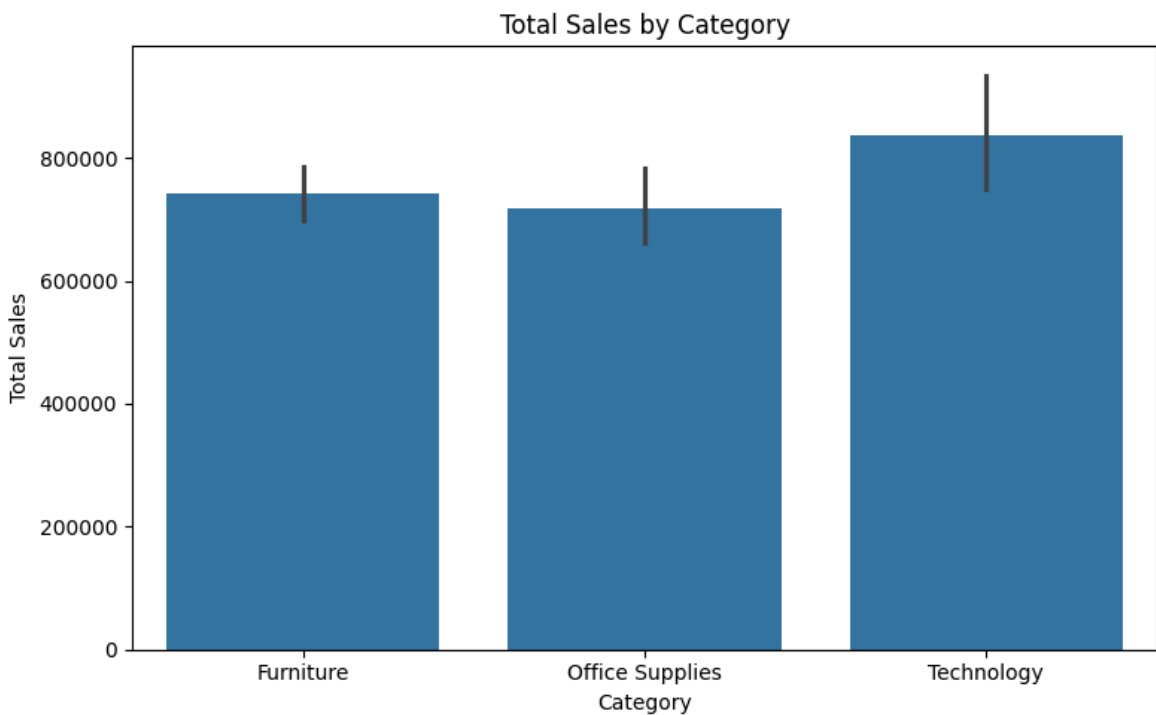
Missing Values:

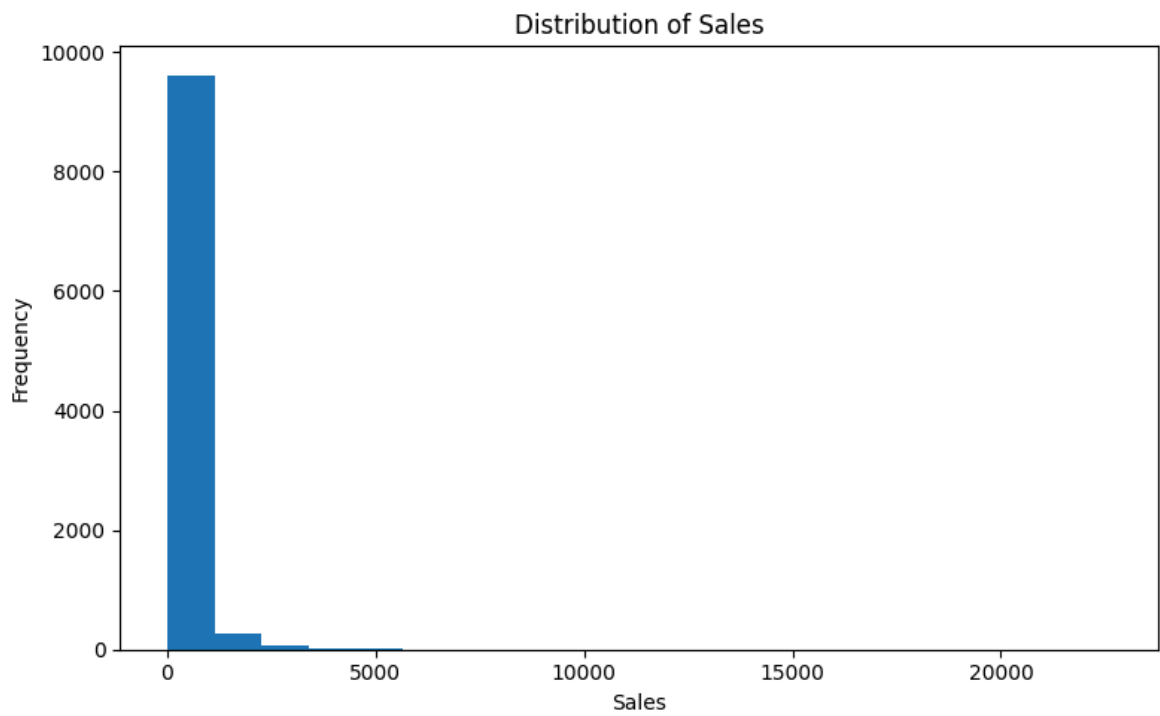
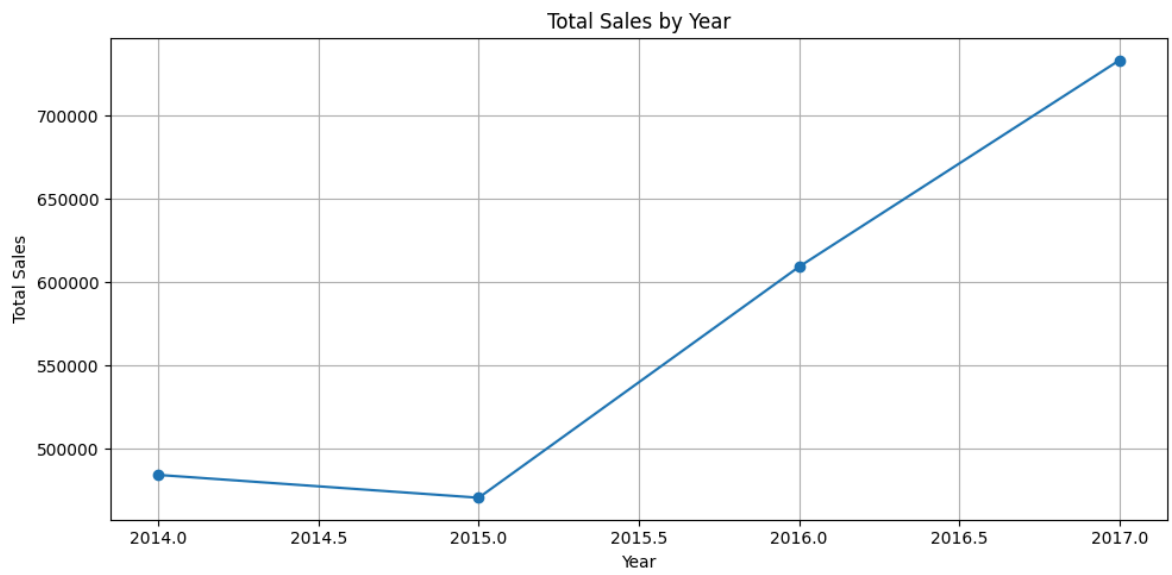
```

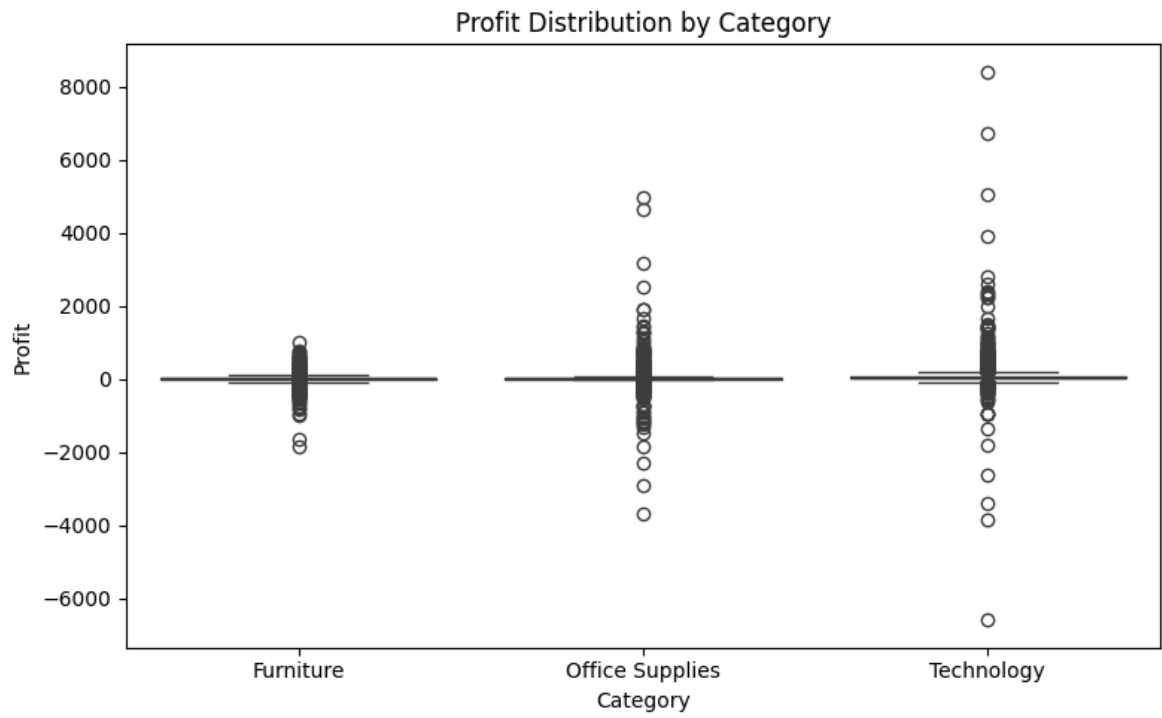
Row ID      0
Order ID    0
Order Date  0
Ship Date   0
Ship Mode   0
Customer ID 0
Customer Name 0
Segment     0
Country     0
City        0
State       0
Postal Code 0
Region      0
Product ID  0
Category    0
Sub-Category 0
Product Name 0
Sales       0
Quantity    0
Discount    0
Profit      0

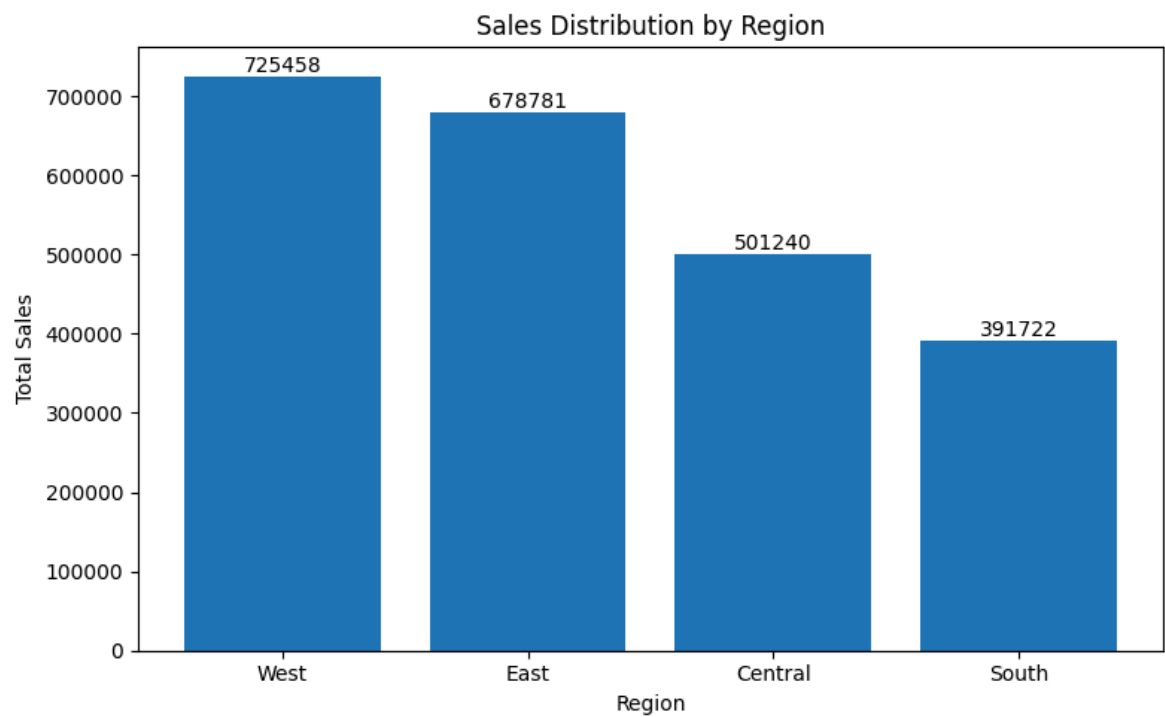
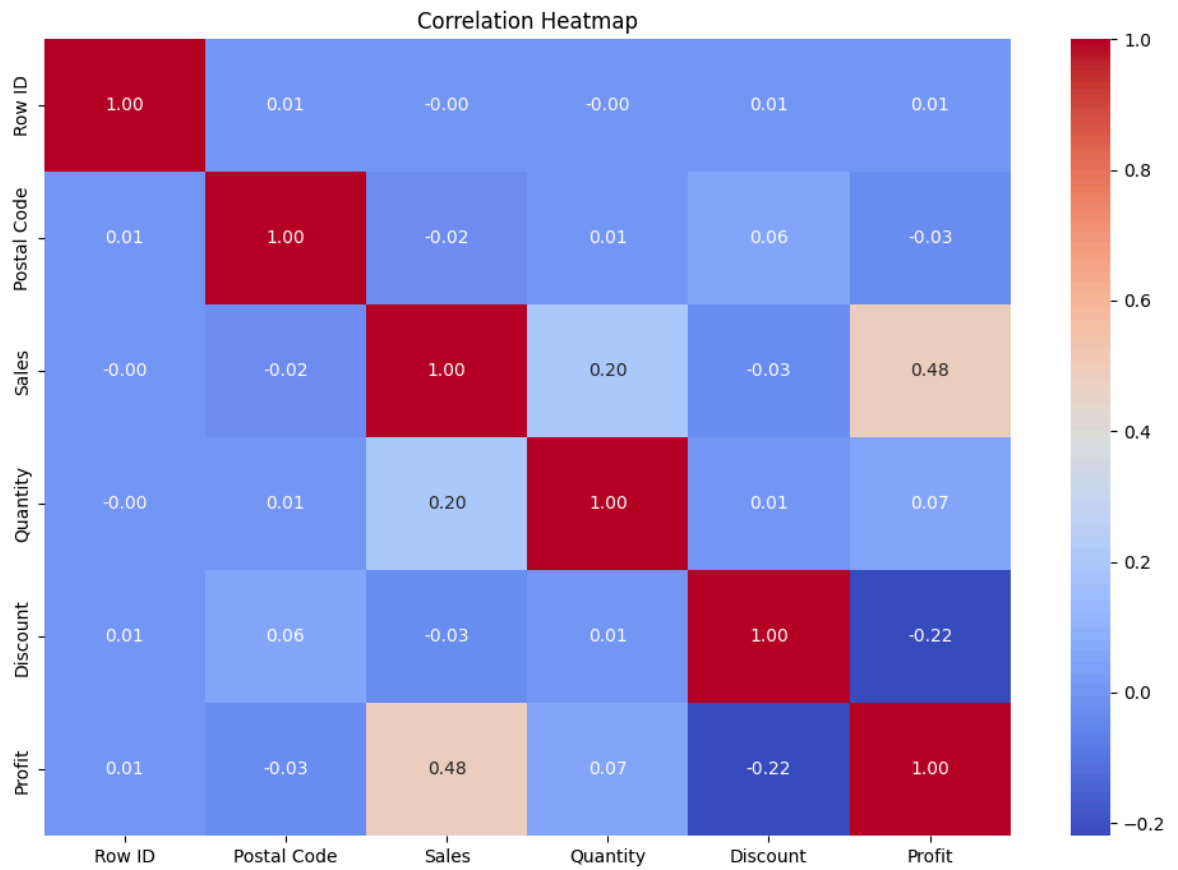
```

dtype: int64









Visualization Summary

1. The bar chart shows total sales for each product category.
2. The line chart shows sales trends across different years.
3. The histogram shows the distribution of sales values.
4. The box plot compares profit distribution across categories.
5. The scatter plot shows the relationship between sales and profit.
6. The heatmap shows correlations between numerical variables.
7. The customized chart compares sales across different regions.

Business Insights

Total Sales: 2297200.86
Total Profit: 286397.02
Total Orders: 5009
Total Customers: 793
Best Category: Technology
Best Region: West

Experiment completed successfully!